

N4L1 Linear Regression:- $(i = 1, \dots, n)$ Given data $\{x_1, y_1, \dots, (x_n, y_n)\}$, $x_i \in \mathbb{R}^d, y_i \in \mathbb{R}$

$$L(\theta) = \frac{1}{2} \sum_{i=1}^n (x_i^T \theta - y_i)^2$$

Minimize L : Define $A = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} \leftarrow \text{Feature Matrix}$, $y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$

$$A\theta = \begin{bmatrix} x_1^T \theta \\ \vdots \\ x_n^T \theta \end{bmatrix}$$

$$A\theta - y = \begin{bmatrix} x_1^T \theta - y_1 \\ \vdots \\ x_n^T \theta - y_n \end{bmatrix}$$

$$(A\theta - y)^T (A\theta - y) = \sum_{i=1}^n (x_i^T \theta - y_i)^2$$

✓ So, $L(\theta) = \frac{1}{2} (A\theta - y)^T (A\theta - y)$

Minimizing L : $\nabla_{\theta} L(\theta) = 0$

$$\nabla_{\theta} ((A\theta - y)^T (A\theta - y)) = 0$$

$$\Rightarrow A^T (A\theta - y) = 0$$

$$\Rightarrow \boxed{(A^T A) \theta = A^T y} \rightarrow \text{Least Square Solution}$$

Note: Can we write $\theta = (A^T A)^{-1} A^T y$?

✓ Yes, if A is full rank [Why? Full rank $A \Rightarrow A^T A$ is invertible]
How to argue this? Show that $\text{rank}(A) = \text{rank}(A^T A)$

By showing $N(A) = N(A^T A)$

(Take an $x \in N(A)$ to show $x \in N(A^T A)$ and the other way)

Max Likelihood and least squares:

Suppose the data is generated according to following model:

$$x \rightarrow \boxed{\text{Linear Model}} \rightarrow y = \theta^T x + \epsilon \quad \rightarrow (\text{zero-mean noise})$$

eg. $\epsilon \sim \text{Gaussian with mean 0 and variance } \left(\frac{1}{\beta}\right)$
 (Precision Parameter)

Dataset $D = \{(x_i, y_i), i = 1 \dots n\}$ generated in an iid fashion.
 (Independent & identically distributed)

Max Likelihood approach is

$$L(\theta) = \prod_{i=1}^n \frac{\sqrt{\beta}}{\sqrt{2\pi}} \exp\left(-\frac{\beta}{2} (y_i - \theta^T x_i)^2\right)$$

Why this function?

Module 6 on Probability.

Max Likelihood approach is to find a θ that maximizes

Instead of $\max_{\theta} L(\theta)$, we could consider $\max_{\theta} \log L(\theta)$

Since log is increasing, the optima of these 2 problems are the same.

$$\log L(\theta) = \frac{n}{2} \log \beta - \frac{n}{2} \log 2\pi - \beta \left[\frac{1}{2} \sum (y_i - \theta^T x_i)^2 \right]$$

(is the least squares objective that we minimized w.r.t. θ)

Message:- "Maximizing $\log L(\theta)$ " is the same as

$$\min_{\theta} \frac{1}{2} \sum (y_i - \theta^T x_i)^2$$

(or)

✓ Least squares regression solves a max Likelihood estimation problem under a linear model.

Polynomial Regression:-

Consider 1-D data

$$D = \{(x_i, y_i), i = 1, \dots, n\} \quad x_i, y_i \in \mathbb{R}, \forall i$$

Previously, we tried fitting a line through this data.

Here, we generalise to any polynomial, say of degree m

Transform Features:-

$$\begin{aligned} \hat{y}(x) &= \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_m x^m \\ &= \sum_{j=0}^m \theta_j \phi_j(x), \text{ where } \phi_j(x) = x^j \end{aligned}$$

For a given x , the transformed feature vector

$$\phi(x) = (1, x, x^2, \dots, x^m)$$

$$\hat{y}(x) = \theta^T \phi(x), \quad \theta = (\theta_0, \dots, \theta_m)$$

Using these transformed features, perform linear regression

$$A = \begin{bmatrix} \phi(x_1)^T \\ \vdots \\ \phi(x_n)^T \end{bmatrix}, \quad Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

AQ 4.

1) For a L.R problem $A\theta = Y$, with A, Y as in lecture notes, which of the following is the correct eqn of the loss function?

$$L(\theta) = \frac{1}{2} (A\theta - Y)^T (A\theta - Y)$$

2) For a L.R problem $A\theta = Y$, the least sq. solⁿ $\theta = (A^T A)^{-1} A^T Y$ exists if

A is full rank.

3) Least sq. regression solves a max likelihood estimation problem under a linear model.

True.

4) For $\lambda > 0$, $(A^T A + \lambda I)$ is always invertible.

True

W4L2 EIGEN Values and EIGEN Vectors

Motivation:- Ordinary Differential Equations

⊗

$$\frac{dv}{dt} = 4v - 5w, \quad v = 8 \text{ at } t=0, \quad \frac{dw}{dt} = 2v - 3w$$

$$w = 5 \text{ at } t=0$$

$$u(t) = \begin{bmatrix} v(t) \\ w(t) \end{bmatrix} \quad u(0) = \begin{bmatrix} 8 \\ 5 \end{bmatrix}$$

* can be written as

$$\frac{du}{dt} = Au, \text{ where } A = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix} \text{ and } u = u(0) \text{ at } t=0$$

Simple Case of Single Eqn:-

$$\frac{du}{dt} = au \quad \text{--- } (**)$$

Soln: $u(t) = e^{at} u(0)$ $\leftarrow$ easy to see that this is a solⁿ of $**$.

$**$ is unstable if $a > 0$ ($e^{at} \rightarrow \infty, t \rightarrow \infty$)

$**$ $\begin{cases} \text{is unstable if } a > 0 \\ \text{is neutrally stable if } a = 0 \\ \text{is stable if } a < 0 \end{cases}$

Case of a 2 D Parameter $\rightarrow$ Want solⁿs of this form following for the system $(*)$:

$$v(t) = e^{\lambda t} y, \quad w(t) = e^{\lambda t} z$$

$$(or) \quad u(t) = e^{\lambda t} x, \quad \text{where } x = \begin{bmatrix} y \\ z \end{bmatrix}$$

If we substitute this in $\frac{du}{dt} = Au$, we obtain

$$\lambda e^{\lambda t} y = 4e^{\lambda t} y - 5e^{\lambda t} z$$

$$\lambda e^{\lambda t} z = 2e^{\lambda t} y - 3e^{\lambda t} z$$

Take out $e^{\lambda t}$, we get

$$\begin{cases} 4y - 5z = \lambda y \\ 2y - 3z = \lambda z \end{cases}$$

$$\Leftrightarrow Ax = \lambda x \rightarrow \text{Eigen Value Eqn}$$

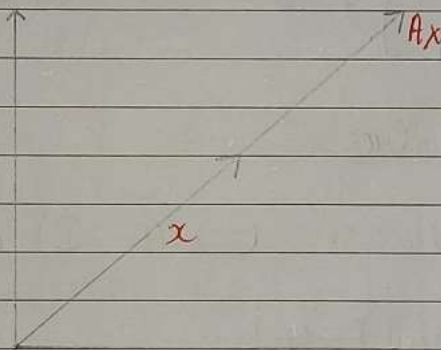
Eigen Value

Eigen Vector
(non trivial
 $x \neq 0$)

Message: Can solve $\frac{du}{dt} = Au$ using solutions of the form $u(t) = e^{\lambda t} x$ if $Ax = \lambda x$ can be solved.

Eigenvalues & Eigenvectors:-

For a matrix A , λ is an eigenvalue & $x \neq 0$ is an eigenvector if $Ax = \lambda x$



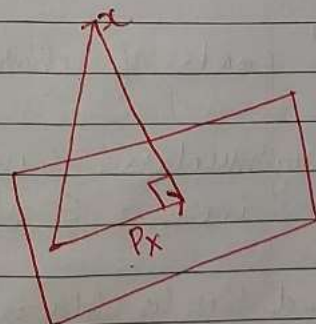
If x is an eigenvector if $Ax = \lambda x$

If x is an eigenvector then, A either stretches it or shrinks it, but A does not change the direction of x .

✓ Case $\lambda = 0$: $Ax = 0 \Rightarrow x \in N(A)$

examples:- ① Projection Matrix P
eg. Project onto a plane

$Px = x$ for any x in plane



$\Rightarrow \lambda = 1$ is an eigenvalue & any x in the plane is an eigenvector

ii) Consider an x that is perpendicular to the plane.

$$Px = 0$$

So, $\lambda = 0$ is another eigenvector & any $x \perp$ plane is eigenvector.

(2) Permutation Matrix:- Take the identity matrix and shuffle its rows.

eg. $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$$Bx = x \text{ for } x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, Bx = -x \text{ for } x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Finding the Eigen Values:-

$$Ax = \lambda x \Rightarrow (A - \lambda I)x = 0 \Rightarrow (A - \lambda I) \text{ is singular}$$

$$\Rightarrow \boxed{\det(A - \lambda I) = 0}$$

(characteristic polynomial of a matrix A)

If A is $n \times n$, then the characteristic polynomial is of degree n .

$$(a_{11} - \lambda) \dots (a_{nn} - \lambda) = 0$$

" n " roots of characteristic polynomial = eigenvalues.

For eigenvectors: Given λ , want $(A - \lambda I)x = 0$
i.e., $x \in N(A - \lambda I)$

(find this to obtain eigen vector corresponding to eigenvalue λ)
(Can use Gaussian elimination)

eg.) $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

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$$\det(A - \lambda I) = 0$$

$$\lambda^2 - 6\lambda + 8 = 0$$

$$\lambda_1 + \lambda_2 = 6, \quad \lambda_1 \lambda_2 = 8 \quad (\text{Check: } \lambda_1 + \lambda_2 = \text{trace}(A))$$

$$\lambda_1 \lambda_2 = \det(A) = \text{prod. of eigenvalues}$$

Eigenvalues: $\lambda_1 = 4, \lambda_2 = 2$

$$A - 4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$(A - 4I)x = 0 \quad \text{for } x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$(A - 2I)x = 0 \quad \text{for } x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

These eigenvectors are linearly independent

Variation: Suppose $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $B + 3I = A$

$$Ax = (B + 3I)x = \lambda x + 3x = (\lambda + 3)x$$

How? $\rightarrow$ Eigenvectors of $B =$ Eigenvectors of A

Remarks: ① Suppose $Ax = \lambda_1 x$ and $Bx = \lambda_2 x$

Then, can we claim $(A+B)x \stackrel{?}{=} (\lambda_1 + \lambda_2)x$

Ans $\rightarrow$ No: Since the x 's need not be the same for $A+B$ i.e. $Ax_1 = \lambda_1 x_1$ and $Bx_2 = \lambda_2 x_2$ with x_1 not necessarily equal to x_2

② Symmetric Matrix has "real" eigenvalues.
A matrix whose eigenvectors are not real: $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$$\det(A - \lambda I) = \lambda^2 + 1$$

$$\lambda_1 = i, \lambda_2 = -i$$

③ A matrix where we don't get "independent" eigenvectors

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} \quad \lambda_1 = \lambda_2 = 3$$

$$(A - \lambda I) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \left(x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \text{ is a eigenvector}$$

and there is no other eigenvector that is linearly independent of x)

AQ 4.2 ① If an eigen vector of matrix A is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, then its corresponding eigenvector belongs to.
Null space of A

② For a symm. Matrix
The eigen values are always real

③ For a matrix A with eigenvalue λ , corresponding eigenvector lies in?
Null space of $A - \lambda I$

④ Consider $A = aI + B$ where A and B are matrices and a is a scalar. True is?
Eigen vectors of A are same as Eigen vectors of B

⑤ For a matrix A having eigenvalue λ , $A - \lambda I$ is a singular matrix.

⑥ If x is an eigenvector of A corresponding to eigenvalue λ_1 and x is also an eigenvector of B corresponding to eigenvalue λ_2 , then x is also an eigenvector of $(A + B)$
False ! True

⑦ A permutation matrix has eigenvalue
1 or -1

Similarity and Diagonalisation:-

Def:- A matrix A is diagonalizable if $\exists$ an invertible matrix S such that.

$$S^{-1}AS = \Lambda$$

→ diagonal matrix

"Matrix being diagonalizable is tied to having enough independent eigen vectors"

Suppose A is an $n \times n$ matrix with n linearly independent eigen vectors, say $\{x_1, \dots, x_n\}$ with corresponding eigen values $\{\lambda_1, \dots, \lambda_n\}$. Then, A is diagonalizable. Why?

Let $S = \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix}$ Is Invertible?
 Yes because $\text{rank}(S) = n$ because $\{x_1, \dots, x_n\}$ is a linearly independent set.

$$\begin{aligned} AS &= A \begin{bmatrix} | & | & & | \\ x_1 & \dots & x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ Ax_1 & \dots & Ax_n \\ | & | & & | \end{bmatrix} \\ &= \begin{bmatrix} | & | & & | \\ \lambda_1 x_1 & \dots & \lambda_n x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ x_1 & \dots & x_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \ddots & & \\ & & \lambda_n & \\ & & & 0 \end{bmatrix} \\ &= S\Lambda, \text{ where } \Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \ddots & & \\ & & \lambda_n & \\ & & & 0 \end{bmatrix} \end{aligned}$$

$$\text{So, } AS = S\Lambda \Leftrightarrow S^{-1}AS = \Lambda$$

Claim: If λ_1, λ_2 are the eigenvalues with corresponding eigenvector x_1, x_2 and $\lambda_1 \neq \lambda_2$. Then $\{x_1, x_2\}$ is a linearly independent set.

Proof: Suppose $c_1 x_1 + c_2 x_2 = 0$ — (1)

$$A(c_1 x_1 + c_2 x_2) = 0$$

$$c_1 \lambda_1 x_1 + c_2 \lambda_2 x_2 = 0 \quad \text{--- (2)}$$

$$\boxed{c_1(\lambda_1 - \lambda_2)x_1 = 0} \quad ((2) - \lambda_2(1))$$

$$\lambda_1 \neq \lambda_2, x_1 \neq 0 \Rightarrow c_1 = 0$$

$$\text{Similarly, } c_2 = 0$$

$\Rightarrow \{x_1, x_2\}$ is a linearly independent set

Extension of this claim: If $\lambda_1, \lambda_2, \dots, \lambda_n$ are distinct, then their eigenvectors $\{x_1, \dots, x_n\}$ are linearly independent.

eg. $\rightarrow A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ (Find eigen values & vectors)

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}$$

$$\text{Characteristic eqn: } \lambda^2 - 4\lambda - 5 = 0$$

$$\text{Roots are: } \lambda_1 = 5, \lambda_2 = -1$$

Q $\rightarrow$ Is A diagonalizable?

Yes, because it has 2 distinct eigen values.

Finding eigen vectors: $\lambda_1 = 5$

$$A - \lambda_1 I = \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \in N(A - \lambda_1 I)$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\textcircled{2} \lambda_2 = -1$$

$$A - \lambda_2 I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ 1 \end{bmatrix} \in N(A - \lambda_2 I), \quad x_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\text{Check: } S^{-1}AS = \Lambda$$

Remarks:

$$\textcircled{1} S^{-1}AS = \Lambda \quad \text{or} \quad A = SAS^{-1}$$

Is S Unique?

No, If x_1 is an eigenvector, then so is cx_1 .
So, the columns of S can be scaled to get S' such that $S'^{-1}AS' = \Lambda$

$$\textcircled{2} A = SAS^{-1} \quad \text{Suppose col. 1 of } S \text{ is } y$$

$$\text{Col. 1 of } S\Lambda = \lambda_1 y,$$

$$\text{Col 1 of } AS = Ay.$$

$$\text{We know } AS = S\Lambda$$

$$Ay = \lambda_1 y \Rightarrow \lambda_1 \text{ is an eigenvalue \& } y \text{ is an eigenvector of } A.$$

$\textcircled{3}$ Powers of A : Suppose λ is an eigenvalue and x is an eigenvector of A .

$$A^2x = A(Ax) = \lambda Ax = \lambda^2 x$$

$$\text{Suppose } S^{-1}AS = \Lambda, \quad \text{Q} \rightarrow \text{Is } S^{-1}A^2S = \Lambda^2? \\ \text{Yes}$$

$$(S^{-1}AS)(S^{-1}AS) = (\Lambda)(\Lambda)$$

$$S^{-1}A^2S = \Lambda^2$$

The argument above works for a general $k \geq 1$, i.e.

$$S^{-1}A^kS = \Lambda^k$$

④ "Not all matrices are diagonalizable"

eg:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Check: A does'nt have 2 linearly independent eigen vectors.

AQ 4.3 1) All matrices are diagonalizable.

False

2) If the eigenvalues of a matrix are distinct, then its eigen vectors are linearly independent.

True

3) Consider a matrix A that satisfies $S^{-1}AS = \Lambda$ then,

1) S contains eigenvectors of A

2) Λ " eigenvalues of A

Which is True.

Both are True

4) If $S^{-1}AS = \Lambda$, then S is unique

False

5) If $S^{-1}AS = \Lambda$ then true is:-

$$S^{-1}A^kS = \Lambda^k$$

W 4.4 Solving Fibonacci Sequence using diagonalization

$$F_k = 0, 1, 1, 2, 5, \dots$$

$$F_{k+2} = F_{k+1} + F_k$$

Q → What is F_{100} ?

A linear algebra solⁿ to this problem uses diagonaliza

System of equations: $F_{k+2} = F_{k+1} + F_k$

$$F_{k+1} = F_{k+1}$$

→ (linear funct.)

$$U_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}, \quad U_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} U_k$$

$$\Leftrightarrow \begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = \begin{bmatrix} F_{k+1} + F_k \\ F_{k+1} \end{bmatrix}$$

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad U_0 \xrightarrow{A} U_1 \xrightarrow{A} U_2 \xrightarrow{A} U_3 \longrightarrow \dots$$

$$U_k = A^k U_0 \text{ is a sol}^n \text{ to } U_{k+1} = A U_k$$

In general scenario, where A is $n \times n$, U_k is $n \times 1$

If A is diagonalizable, i.e. A has " n " independent eigen vectors, the

$$U_0 = C_1 x_1 + C_2 x_2 + \dots + C_n x_n$$

Since $\{x_1, \dots, x_n\}$ is a basis.

$$\begin{aligned} A U_0 &= C_1 A x_1 + \dots + C_n A x_n \\ &= C_1 \lambda_1 x_1 + \dots + C_n \lambda_n x_n \end{aligned}$$

$$A^2 u_0 = A(c_1 \lambda_1 x_1 + \dots + c_n \lambda_n x_n) \\ = c_1 \lambda_1^2 x_1 + \dots + c_n \lambda_n^2 x_n$$

$$u_k = A^k u_0 = c_1 \lambda_1^k x_1 + \dots + c_n \lambda_n^k x_n$$

Back to Fibonacci: to find u_k , we need to check if $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ is diagonalizable

Δ find its eigenvectors if yes.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{characteristic eqn: } \lambda^2 - \lambda - 1 = 0$$

$$\lambda_1 = \frac{1 + \sqrt{5}}{2}, \quad \lambda_2 = \frac{1 - \sqrt{5}}{2}$$

$$u_0 = \begin{bmatrix} F_1 \\ F_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 x_1 + c_2 x_2$$

Let's find eigenvectors of A :

$$(A - \lambda I)x = \begin{bmatrix} 1 - \lambda & 1 \\ 1 & -\lambda \end{bmatrix} x = 0$$

$$x_1 = \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}$$

$$(A - \lambda_1 I) \begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix} = \begin{bmatrix} \lambda_1^2 - \lambda_1 - 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{Since } \lambda_1 \text{ is root of } \lambda^2 - \lambda - 1 = 0$$

$$\text{Similarly } (A - \lambda_2 I) \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix} = 0$$

So, x_1 and x_2 are eigenvectors corresponding to λ_1 and λ_2 writing u_0 as a linear combo of x_1, x_2 .

$$U_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{bmatrix}$$

$$\Rightarrow c_1 = \frac{1}{\sqrt{5}}, c_2 = -\frac{1}{\sqrt{5}}$$

$$U_k = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2$$

$$\begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^k \begin{bmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{bmatrix} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^k \begin{bmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{bmatrix}$$

$$F_k = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^k - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^k$$

as k increases $\left(\frac{1-\sqrt{5}}{2} \right)^k$ becomes

negligible. A good enough approx. $F_k \approx \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^k$

(Or)

$$F_{100} \approx \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{100} = \frac{1}{\sqrt{5}} (1.618)^{100}$$

Bottomline:- Diagonalizability can be used to understand linear recurrence relations.

AQ 4.4 1) Which of the foll. ^{are} of eigenvalues of matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ and (2) eigen vectors of A .

①

Eigen Value

②

Eigen Vector

and

$$\left[\frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2} \right]$$

$$\begin{bmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{bmatrix}, \begin{bmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{bmatrix}$$

W4B5 Orthogonally Diagonalizable Matrices:

Main Result: A is a real symmetric matrix. Then,

- 1) Eigenvalues of A are real
- 2) Eigenvectors corresponding to different eigen values are linearly independent.
- 3) A is orthogonally diagonalizable

(This means)

$\exists Q$ satisfy $Q^T Q = I$ such that $Q^{-1} = Q^T$

$$A = Q \Lambda Q^T$$

$$= \begin{bmatrix} | & & | \\ x_1 & \dots & x_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} \text{---} x_1^T \text{---} \\ \text{---} x_2^T \text{---} \\ \text{---} x_n^T \text{---} \end{bmatrix}$$

Remark ① A is diagonalizable + Orthogonal matrix for diagonalization
 $(A = Q \Lambda Q^{-1})$ $(Q^T Q = I)$

$\swarrow$
 (orthogonally diagonalizable)
 $(A = Q \Lambda Q^T)$

② Any real matrix is not necessarily diagonalizable, but a real symmetric is diagonalizable.

eg. $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is not diagonalizable

eg. $\rightarrow A = \begin{bmatrix} 1 & -2 \\ -2 & -2 \end{bmatrix}$

Eigen Values: $\lambda_1 = -3, \lambda_2 = 2$

Eigen Vectors: $x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, x_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$$q_1 = \frac{x_1}{\|x_1\|} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$q_2 = \frac{x_2}{\|x_2\|} = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} | & | \\ q_1 & q_2 \\ | & | \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} \quad \text{Check } Q^T Q = I$$

$$\text{Then, } Q A Q^T = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 \\ -2 & -2 \end{bmatrix} = A$$

AQ 4.5 ① A matrix P is orthogonal if
 $P^T = P^{-1}$

② Suppose that A and P are 2×2 matrices and P is invertible.
 If $P^{-1} A P = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, then eigenvalues of matrix A are

Difference w.r.t regular linear regression that we are using transformed features and then performing regression

Regularised Version of Linear Regression (A.K.A Ridge Regression)

Instead of solving $\min_{\theta} \frac{1}{2} \sum_{i=1}^n (x_i^T \theta - y_i)^2$, we solve the following regularised version

$$\bar{L}(\theta) = \frac{1}{2} \sum_{i=1}^n (x_i^T \theta - y_i)^2 + \underbrace{\lambda \|\theta\|^2}_{\text{regularisation term}}$$

Advantage:- Repeat the calculation leading to least squares solⁿ, we obtain

$$(A^T A + \lambda I) \theta_{\text{reg}} = A^T Y$$

$$\theta_{\text{reg}} = (A^T A + \lambda I)^{-1} A^T Y$$

✓ Show $(A^T A + \lambda I)$ is invertible even if A is not full rank

Note:- Parameter λ controls overfitting

Too small λ leads $\rightarrow$ overfitting

// Large λ leads $\rightarrow$ underfitting

Notes By Hardik Sharma

If any part is blurred or not understandable
message me on WhatsApp (9877100914)

Tip: Don't directly jump onto the notes firstly
watch the whole video and then read them.